

(1001CJA102120110)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
10-07-2021
JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A	D	A,B,C,D	B,C,D	B,C	A,C,D	B,C,D	B
	Q.	11	12								
	A.	A,B,C	B,C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	1.00	1.60	0.07 to 0.08	12.00	2.23 to 2.24	24.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	A	C	A,C	C,D	A,B,C,D	A,B,C	C,D	B,D
	Q.	11	12								
	A.	A,B,C	B,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	374.15	0.06 or 0.07	158.00	6.00	2.00	5.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	B	B	B,C	A,B	B,C	B,C,D	B,C,D	A,B,C,D
	Q.	11	12								
	A.	B,C	B,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	0.37	0.66 or 0.67	1.73	1.50	4.00	1.70				

(1001CJA102120111)

Test Pattern

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JEE(Advanced)
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10-07-2021
JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,B,D	B,C	D	A,B,D	C,D	B,D	A,B,D	A	C
	Q.	11	12								
	A.	C	D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	1.10 to 1.15	5.00	960.00	22.22	0.40	7.35 to 7.41				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	D	A,B,C	A,B,D	A,C,D	C	A,C,D	D	B
	Q.	11	12								
	A.	A	C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	5.50	-0.90	1650.00	5.00	3.00	3.00				

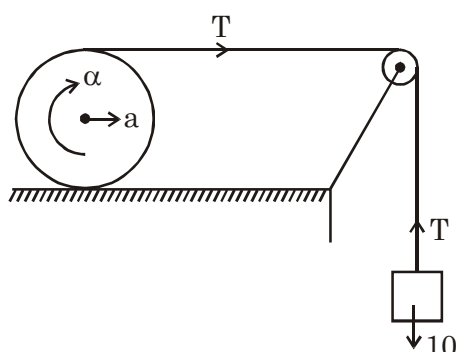
PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	C	A,B,D	A,C	A,C	B,C,D	A,C	B,C	A	B
	Q.	11	12								
	A.	A	B								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	4.00	5.00	15.00	5.00	7.00	6.00				

PAPER-1**PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (C)

Sol. Smooth



$$10 - T = 1a_1 = a + R\alpha$$

$$TR = \frac{1}{2} \times 2R^2\alpha = R\alpha$$

$$T = 2a$$

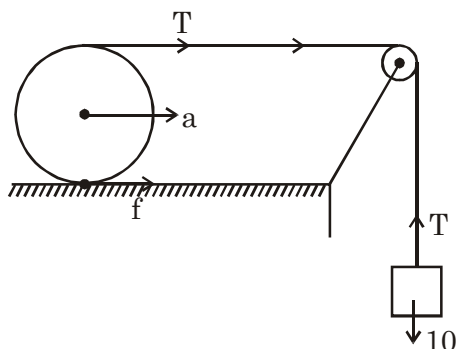
$$\Rightarrow 10 - T = \frac{T}{2} + T$$

$$10 = 2.5 T$$

$$\Rightarrow T = 4$$

$$a = 2 \text{ m/s}^2 \Rightarrow \ell = \frac{1}{2} \times 2t_1^2 \quad t_1 = \sqrt{\ell}$$

Case (ii)



$$10 - T' = 1a'_1 = a' + R\alpha' = 2R\alpha'$$

$$T' \times 2R = \frac{3}{2} \times 2 \times R^2\alpha'$$

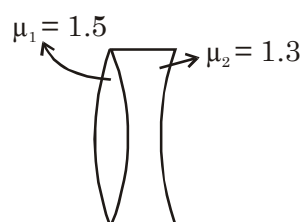
$$R\alpha' = \frac{2T'}{3}$$

$$10 - T' = 2 \times \frac{2T'}{3}$$

$$10 = \frac{7T'}{3} \Rightarrow T' = \frac{30}{7} \Rightarrow a' = \frac{20}{7}$$

$$\ell = \frac{1}{2} \times \frac{20}{7} \times t_2^2 \Rightarrow t_2 = \sqrt{\frac{7\ell}{10}}$$

2. Ans. (B)



Sol.

 $F_{eq} \Rightarrow +ve$ converging. $F_{eq} \Rightarrow -ve$ diverging.

$$\frac{1}{F_{eq}} = \frac{1}{F_1} + \frac{1}{F_2}$$

$$\frac{1}{F_1} = \left[\frac{\mu_1}{\mu_m} - 1 \right] \left[\frac{2}{R} \right] \quad \text{and} \quad \frac{1}{F_2} = \left[\frac{\mu_2}{\mu_m} - 1 \right] \left[\frac{-2}{R} \right]$$

$$\frac{1}{F_{eq}} = \frac{2}{R} \left[\left(\frac{\mu_1}{\mu_m} - 1 \right) - \left(\frac{\mu_2}{\mu_m} - 1 \right) \right]$$

$$\frac{1}{F_{eq}} = \frac{2}{R} \left[\frac{\mu_1}{\mu_m} - \frac{\mu_2}{\mu_m} \right]$$

$$(A) \quad \frac{1}{F_{eq}} = \frac{2}{R} \left[\frac{1.5}{1.4} - \frac{1.3}{1.4} \right], \quad \frac{1}{F_{eq}} = \frac{2}{R} \left[\frac{1.5}{1.6} - \frac{1.3}{1.6} \right]$$

Converging

Converging

$$(B) \quad \frac{1}{F_{eq}} = \frac{2}{R} \left[\frac{1.5}{1.2} - \frac{1.3}{1.2} \right], \quad \frac{1}{F_{eq}} = \frac{2}{R} \left[\frac{1.5}{1.4} - \frac{1.3}{1.4} \right]$$

Converging

Converging

$$(C) \quad \frac{1}{F_{eq}} = \frac{2}{R} \left[\frac{1.5}{1.6} - \frac{1.3}{1.6} \right], \quad \frac{1}{F_{eq}} = \frac{2}{R} \left[\frac{1.5}{1.2} - \frac{1.3}{1.2} \right]$$

Converging Converging

3. Ans. (A)

Sol. $\Delta W_{AB} + \Delta U_{AB} = 0$

$$\Delta U_{AB} = -\frac{1}{2} \times 3 \times 10^5 (10^{-3}) = -150 \text{ J}$$

$$\Delta Q_{ACB} = W + \Delta V$$
$$= -150 + 200 = 50 \text{ J}$$

4. Ans. (D)

Sol. $q = 10\mu\text{C} = 10^{-5}\text{C}$

$$m = 10^{-3} \text{ kg}$$

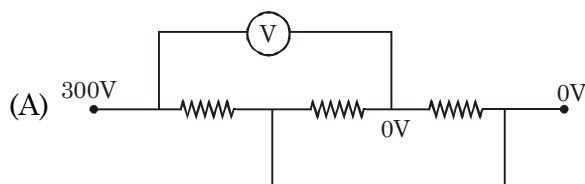
$$\vec{v} = 10^6 \left[\frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{j} \right]$$

$$\vec{F} = q[\vec{v} \times \vec{B}]$$

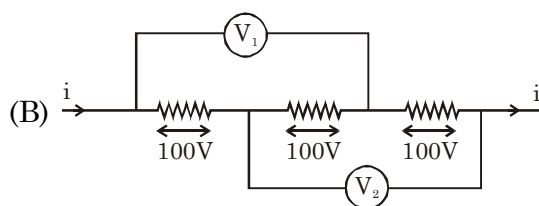
5. Ans. (A,B,C,D)

Sol. Ideal voltmeter ($R_V = \infty$)

Ideal ammeter ($R_A = 0$)



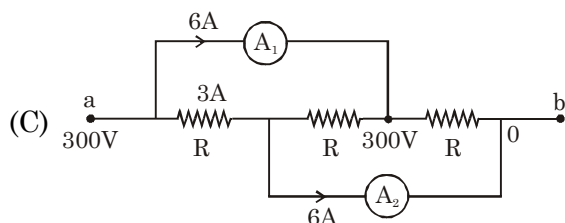
Reading = 300V.



$$i = \frac{300}{300} = 1A$$

$$V_1 = 100 + 100 = 200V$$

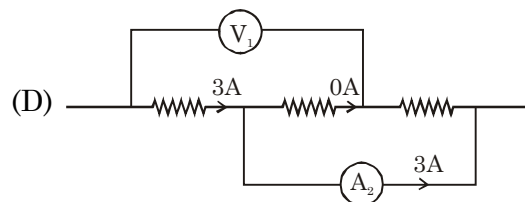
$$V_2 = 100 + 100 = 200V$$



All resistance are in parallel combination.

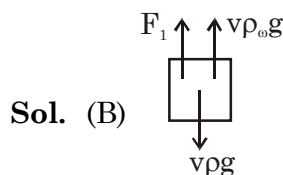
$$i = \frac{300}{100/3} = 9A$$

Reading of ammeter is 6A



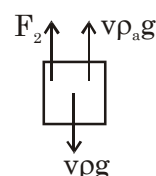
$$i = \frac{300}{100} = 3A$$

6. Ans. (B,C,D)



$$F_1 + v\rho g = v\rho_0 g$$

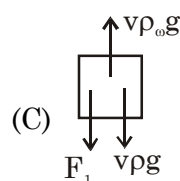
$$F_1 = v (\rho - \rho_w)g$$



$$F_2 = v (\rho - \rho_a)g$$

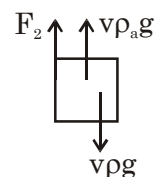
$$\rho_a < \rho_\Theta \Rightarrow F_1 < F_2$$

⇒ easy to hold in water



$$F_1 = v (\rho_0 - \rho)g$$

$$F_2 = v(\rho - \rho_a)g$$



$$\text{if } \rho < \frac{\rho_{\omega} + \rho_a}{2}$$

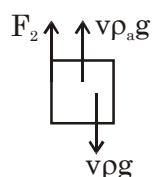
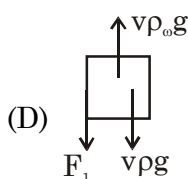
$$F_1 > v \left[\rho_w - \frac{\rho_w + \rho_a}{2} \right] g$$

$$F_1 > v \left[\frac{\rho_w - \rho_a}{2} \right] g$$

$$\text{and } F_2 > v \left[\frac{\rho_w - \rho_a}{2} - \rho_a \right] g$$

$$F_2 < v \left[\frac{\rho_w - \rho_a}{2} \right] g$$

it is easier to hold in air



$$F_1 = v (\rho_w - \rho) g$$

$$F_2 = v (\rho - \rho_a) g$$

$$\text{if } \rho > \frac{\rho_w + \rho_a}{2}$$

$$\Rightarrow F_1 < v \left[\rho_w - \frac{\rho_w + \rho_a}{2} \right] g$$

$$\Rightarrow F_1 < v \left[\rho_w - \frac{\rho_w + \rho_a}{2} \right] g$$

$$\text{and } F_2 > v \left[\frac{\rho_w + \rho_a}{2} - \rho_a \right] g$$

$$F_2 > v \left(\frac{\rho_w - \rho_a}{2} \right) g$$

$\therefore$ It is easier to hold in water

7. **Ans. (B,C)**

Sol. $Q = CB\ell v$

$$\ell = x \tan \alpha$$

$$\Rightarrow Q = CB\ell v x \tan \alpha$$

$$i = CB\ell v^2 \tan \alpha$$

$$F = Bi\ell = Bi \times x \tan \alpha = Bi(vt \tan \alpha + x_0)$$

$$\tau = 0$$

$$U = \frac{Q^2}{2C} = \frac{(CB\ell v(x_0 + vt) \tan \alpha)^2}{2C}$$

8. **Ans. (A,C,D)**

Sol. ${}_{Z_1}A^{A_1} \longrightarrow {}_{Z_1-1}C^{A_1-8}$ and

$${}_{Z_2}B^{A_2} \longrightarrow {}_{Z_2+3}C^{A_2-4}$$

$$\therefore Z_1 - 1 = Z_2 + 3 \Rightarrow Z_1 - Z_2 = 4$$

$$\text{and } A_1 - 8 = A_2 - 4 \Rightarrow A_1 - A_2 = 4$$

At time t

$$N_A = 4N_0 e^{-\lambda_1 t} \text{ and } N_B = N_0 e^{-\lambda_2 t}$$

$$\text{when } N_A = N_B \Rightarrow 4e^{-\lambda_1 t} = e^{-\lambda_2 t}$$

$$\Rightarrow t = \frac{2 \ln 2}{\lambda_1 - \lambda_2} = 4 \text{ hr}$$

$$N_C = 4N_0(1 - e^{-\lambda_1 t}) + N_0(1 - e^{-\lambda_2 t})$$

putting the value to $t = 4 \text{ hr}$, we get

$$N_c = \frac{9N_0}{2}$$

9. **Ans. (B,C,D)**

Sol. If apparatus is filled with ozone

$$f = \frac{v_1}{4\ell} = \frac{1}{4\ell} \sqrt{\frac{\gamma_1 RT}{M_1}} \dots (i)$$

If apparatus is filled with N_2

$$f = \frac{v_2}{4(\ell + 10)} = \frac{1}{4(\ell + 10)} \sqrt{\frac{\gamma_2 RT}{M_2}} \dots (ii)$$

From (i) & (ii)

$$\frac{1}{\ell} \sqrt{\frac{\gamma_1}{M_1}} + \frac{1}{(\ell + 10)} \sqrt{\frac{\gamma_2}{M_2}}$$

$$\frac{1}{\ell} \sqrt{\frac{4/3}{48}} = \frac{1}{(\ell + 10)} \sqrt{\frac{7/5}{28}}$$

$$\frac{1}{6\ell} = \frac{1}{2(\ell + 10)\sqrt{5}}$$

$$6\ell = 4.5(\ell + 10)$$

$$\ell = \frac{45}{1.5} = 30 \text{ cm} = 0.3 \text{ m}$$

(A) Wavelength of sound in ozone = 4ℓ
= 1.2 m

(B) Wavelength of sound in N_2 = $4(\ell + 10)$
= 160 cm = 1.6 m

(C) For second resonance in ozone shifting

of water column = $\frac{\ell}{2}$

$$= \frac{1.2}{2} \text{ m} = 60 \text{ cm}$$

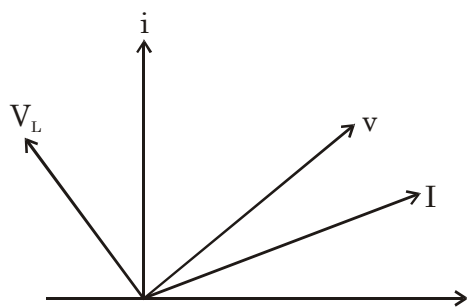
(D) For second resonance in N_2 , shifting

of water column = $\frac{1.2}{2} \text{ m}$

$$= \frac{1.6}{2} \text{ m} = 0.8 \text{ m} = 80 \text{ cm}$$

10. Ans. (B)

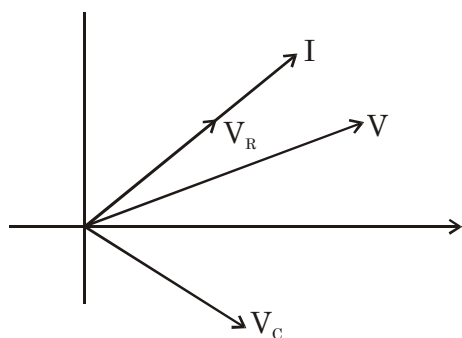
Sol. (i) when box contains inductor, phasor diagram will be



Hence V_L is leading with V of source, So A is not possible.

and V_R is lagging behind V of source, So B is possible.

(ii) When box contains capacitor, phasor diagram will be



Hence V_R is leading with V of source, So C is not possible.

And V_C is lagging behind V , So D is also not possible

11. Ans. (A,B,C)

Sol. $V = \int \vec{F} \cdot d\vec{r} = \frac{kr^2}{2}$

$$\therefore \frac{mv^2}{r} = kr \Rightarrow v^2 \propto r^2 \Rightarrow v \propto r \text{ or } v = cr$$

$$\text{also } mvr = \frac{nh}{2\pi} \Rightarrow cr^2 \propto n \Rightarrow r \propto \sqrt{n} \text{ and also}$$

$$v \propto r \propto \sqrt{n}$$

$$\omega = v/r \propto n^0$$

$$TE = KE + PE$$

$$\propto \frac{1}{2} mV^2 + \frac{1}{2} Kr^2$$

$$TE \propto r^2 \propto (\sqrt{n})^2 \propto n$$

$$\therefore TE \propto n$$

12. Ans. (B,C,D)

SECTION-II

1. Ans. 1.00

Sol. Fringe width,

$$\beta = \frac{D\lambda}{d} = \frac{1}{3} \text{ mm}$$

At a point 1mm above maxima of order-1

= 4 th order maxima will be formed

$$\therefore \text{Intensity at this point} = 1 \text{ W/m}^2$$

2. Ans. 1.60

Sol. $f_1 = f_0$

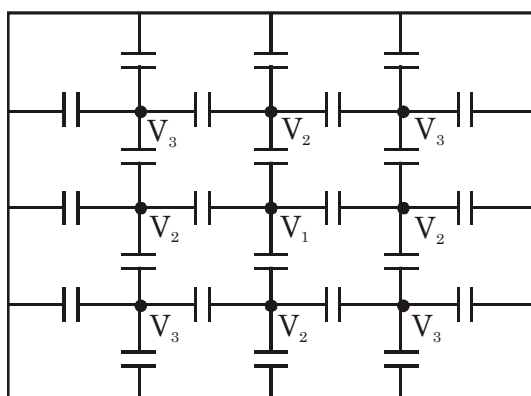
$$f_2 = \frac{c+v}{c-v} \times f_0$$

$$f_b = 256 \left(\frac{2v}{c-v} \right) = 1.6 \text{ Hz}$$

3. Ans. 0.07 to 0.08

4. Ans. 12.00

Sol.



$$2(V_3 - 0) + 2(V_3 - V_2) = 0$$

$$\Rightarrow 2V_3 = V_2$$

$$4(V_1 - V_2) = q$$

$$V_1 - V_2 = \frac{q}{4}$$

$$(V_2 - 0) + (V_2 - V_1) + 2(V_2 - V_3) = 0$$

$$4V_2 = V_1 + 2V_3$$

$$\Rightarrow 3V_2 = V_1 \Rightarrow \frac{2V_1}{3} = \frac{q}{4}$$

$$\Rightarrow V_1 = \frac{3q}{8} = 3V$$

$$V_2 = \frac{V_1}{3} = \frac{q}{8} = 1V$$

$$V_3 = \frac{V_2}{2} = \frac{q}{16} = \frac{1}{2}V$$

$$U = \frac{1}{2} \times C \times V_3^2 \times 8 + \frac{1}{2} \times C \times V_2^2 \times 4 +$$

$$\frac{1}{2} C(V_1 - V_2)^2 \times 4$$

$$+ \frac{1}{2} \times C \times (V_2 - V_3)^2 \times 8$$

$$= \frac{1}{2} \times \frac{1}{4} \times 8 + \frac{1}{2} \times 1^2 \times 4^2 + \frac{1}{2} \times 4 \times 4 + \frac{1}{2} \times \frac{1}{4} \times 8$$

$$= 12 \mu J$$

5. Ans. 2.23 to 2.24

Sol. $\rho = \rho_0 \left(1 - \frac{r^2}{a^2} \right)$

By Gauss law :

$$E(4\pi r^2) = \frac{1}{\epsilon_0} \int_0^r \rho(4\pi r^2) dr$$

$$Er^2 = \frac{\rho_0}{\epsilon_0} \int_0^r \left(1 - \frac{r^2}{a^2} \right) r^2 dr$$

$$\frac{\rho_0}{\epsilon_0} \left[\frac{r^3}{3} - \frac{r^5}{5a^2} \right]$$

for maximum E

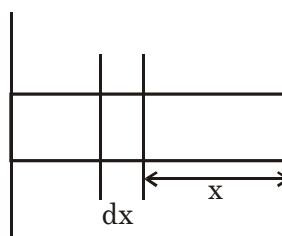
$$\frac{dE}{dr} = 0 \quad \frac{1}{3} - \frac{3r^2}{5a^2} = 0$$

$$\Rightarrow \frac{3r^2}{5a^2} = \frac{1}{3} \Rightarrow r = \sqrt{\frac{5a^2}{9}}$$

$$\Rightarrow r = \sqrt{\frac{5}{9}} a = 2.23 \text{ fm}$$

6. Ans. 24.00

Sol. In steady state,



$$d\ell = dx \propto \Delta T$$

$$T = ax + b$$

$$x = 0, T = 0$$

$$x = 1\text{m}, T = 100^\circ\text{C}$$

$$T = 100x$$

$$\int d\ell = \int 100x \times dx \times 10^{-5}$$

$$\Delta\ell = 10^{-3} \times \frac{1}{2}$$

$$F = \frac{YA}{\ell} \Delta\ell$$

$$= \frac{2 \times 10^{11} \times 4 \times 10^{-6} \times 10^{-3}}{1} \times \frac{1}{2} = 400 \text{ N}$$

$$2\mu N = mg$$

$$m = \frac{2 \times 0.3 \times 400}{10} = 24 \text{ kg}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(B)

2. Ans.(A)

3. Ans.(A)

4. Ans.(C)

$$\text{Solubility, } S = \frac{5520 \times 10^{-6}}{276} = 2 \times 10^{-5} \text{ M}$$

$$K_{sp} = 4S^3 = 4 \times (2 \times 10^{-5})^3 = 3.2 \times 10^{-14}$$

5. Ans.(A,C)

6. Ans.(C,D)

7. Ans.(A,B,C,D)

Tanning of leather, delta formation and cottrell precipitation are applications and examples of colloids.

8. Ans.(A,B,C)

$$\Lambda_m^0(\text{CH}_3\text{COOH}) = \lambda_m^0(\text{H}^+) + \lambda_m^0(\text{CH}_3\text{COO}^-)$$

$$= 3.474 \times 10^{-2} + 1.351 \times 10^{-2}$$

$$= 4.825 \times 10^{-2} \text{ ohm}^{-1}\text{m}^2\text{mol}^{-1}$$

$$k = G \times G^* = \frac{1}{2000} \times 20 = 0.01 \text{ ohm}^{-1}\text{m}^{-1}$$

$$\Lambda_m = \frac{k}{C} = \frac{0.01 \text{ ohm}^{-1}\text{m}^{-1}}{0.1 \times 10^{-3} \text{ molm}^{-3}}$$

$$= 1 \times 10^{-4} \text{ ohm}^{-1}\text{m}^2\text{mol}^{-1}$$

$$\text{Now, } \alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{1 \times 10^{-4}}{4.825 \times 10^{-2}}$$

$$\therefore [\text{H}^+] = C\alpha = 0.1 \times \frac{10^{-2}}{4.825} = 2.07 \times 10^{-4} \text{ M}$$

9. Ans.(C,D)

10. Ans.(B,D)

11. Ans.(A,B,C)

12. Ans.(B,D)

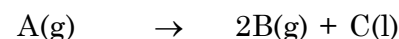
SECTION-II

1. Ans.(374.15)

$$\Delta T_b = K_b \cdot m = 0.52 \times \frac{\frac{11.7}{58.5} \times 2 + \frac{34.2}{34.2}}{0.260} = 1 \text{ K}$$

$$\therefore \text{B.P.} = 373.15 + 1 = 374.15 \text{ K}$$

2. Ans.(0.06 or 0.07)



$$t = 0 \quad 0.08 \text{ bar} \quad 0$$

$$t = 10 \text{ min} \quad (0.08 - x) \text{ bar} \quad 2x \text{ bar}$$

$$t = \infty \quad (\approx 0) \quad 0.16 \text{ bar}$$

but at $t = \infty$, total pressure is 0.165 bar.

hence $(0.165 - 0.160) = 0.005 \text{ bar}$ is the vapour pressure of C(l).

$$\text{Now, at } t = 10 \text{ min, } (0.08 - x) + 2x + 0.005 = 0.125$$

$$\therefore x = 0.04$$

Hence,

$$k = \frac{1}{t} \cdot \ln \frac{P_A^0}{P_A} = \frac{1}{10} \cdot \ln \frac{0.08}{0.08 - 0.04} = 0.06931 \text{ min}^{-1}$$

3. Ans.(158.00)

$$\Delta U = n \cdot C_{v,m} \cdot (T_2 - T_1) = 1 \times 3.2R \times (800 - 300)$$

$$= 3.2 \times 0.08 \times 500 = 128 \text{ L-atm/mol.}$$

$$\text{Now, } \Delta H = \Delta U + \Delta(PV) = 128 + (10 \times 5 - 10 \times 2)$$

$$= 158 \text{ L-atm}$$

4. Ans.(6.00)

Sol. (1,2,3,5,8,9).

5. Ans.(2.00)

6. Ans.(5.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (B)**

Sol.
$$\frac{\sin y}{\cos 3y} = \frac{1}{2} \frac{2 \sin y \cos y}{\cos 3y \cdot \cos y} = \frac{1}{2} \left[\frac{\sin 2y}{\cos 3y \cos y} \right]$$

$$= \frac{1}{2} \left[\frac{\sin(3y - y)}{\cos 3y \cos y} \right] = \frac{1}{2} [\tan 3y - \tan y]$$

$\therefore S = \frac{1}{2} [\tan 3x - \tan x + \tan 3^2 x - \tan 3x$
 $+ \dots + \tan 3^n x - \tan 3^{n-1} x]$

$$= \frac{1}{2} [\tan 3^n x - \tan x]$$

2. **Ans. (D)**

Sol. Diagonals of rhombus are angle bisector of adjacent sides.

Equation of bisector of the sides are

$$\frac{ax + y - 16}{\sqrt{1 + a^2}} = \pm \left(\frac{x + ay - 6}{\sqrt{1 + a^2}} \right)$$

$$\Rightarrow ax + y - 16 = \pm(x + ay - 6)$$

passes through (3,5)

$$\Rightarrow 3a + 5 - 16 = \pm(3 + 5a - 6)$$

$$\Rightarrow a = -4, \frac{7}{4}$$

$$\therefore \text{sum of values} = -\frac{9}{4}$$

3. **Ans. (B)**

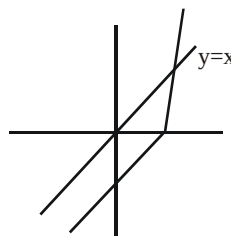
Sol. $\therefore x^3 + y^3 + (-1)^3 = 3(x)(y)(-1)$
 $\Rightarrow x + y + -1 = 0$ or $x = y = -1$
 $\therefore (x, y) \neq (-1, -1)$
 $\therefore$ locus of P is $x + y - 1 = 0$
 $\therefore$ Equation of family of circles touching $x + y - 1 = 0$ at $(-1, 2)$ is
 $(x + 1)^2 + (y - 2)^2 + \lambda(x + y - 1) = 0$
 $\therefore$ It passes through $(1, -2)$
 $\Rightarrow (1 + 1)^2 + (y - 2)^2 + \lambda(1 - 2 - 1) = 0$
 $\Rightarrow 20 - 2\lambda = 0 \Rightarrow \lambda = 10$
 $\therefore$ Circle is $x^2 + y^2 + 12x + 6y - 5 = 0$

4. **Ans. (B)**

Sol. $f(x) = 2x - 1 + \left| x - \frac{1}{2} \right|$

$$= \begin{cases} x - \frac{1}{2} & x < \frac{1}{2} \\ 3x - \frac{3}{2} & x \geq \frac{1}{2} \end{cases}$$

As $f(x)$ is increasing, solution of equation $f(x) = f^{-1}(x)$ is same as $f(x) = x$.



$\therefore$ No. of solutions of $f(x) = f^{-1}(x)$ is one.

5. **Ans. (B,C)**

Sol. Using LMVT, for $x \in (-4, 4)$,

$$\frac{f'(x) - f'(-4)}{x + 4} \geq -3$$

$$\Rightarrow f(x) \geq -2 - 3x \quad \dots(i)$$

$$\text{Also } \frac{f'(4) - f'(x)}{4 - x} \geq -3$$

$$\Rightarrow f'(x) \leq -2 - 3x \quad \dots(ii)$$

from (i) and (ii)

$f'(x) = -2 - 3x \Rightarrow f(x)$ has maximum at

$$x = -\frac{2}{3}$$

$$f(x) = -2x - \frac{3x^2}{2} \quad [\because f(0) = 0]$$

$$g'(x) = f(x) = -2x - \frac{3x^2}{2}$$

$\forall x \in [0, 4) \quad g'(x) < 0 \Rightarrow g(x)$ is decreasing

Also, $g''(x) = f'(x) = -2 - 3x < 0 \quad \forall x \in (0, 4)$

$\Rightarrow$ Graph of $g(x)$ is concave down in $(0, 4)$

6. Ans. (A,B)

$$\begin{aligned}
 \text{Sol. } \int \frac{x^4 + 1}{x^6 + 1} dx &= \int \frac{x^4 - x^2 + 1 + x^2}{(x^2 + 1)(x^4 - x^2 + 1)} dx \\
 &= \int \frac{dx}{x^2 + 1} + \int \frac{x^2}{(x^3)^2 + 1} dx \\
 &= \tan^{-1} x + \frac{1}{3} \tan^{-1}(x^3) + c \\
 \int \frac{(x^2 + 1)^2 - 2x^2}{(x^2 + 1)(x^4 - x^2 + 1)} dx &= \int \frac{x^2 + 1}{x^4 - x^2 + 1} dx - 2 \int \frac{x^2}{(x^3)^2 + 1} dx \\
 &= \int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 1} dx - \frac{2}{3} \int \frac{d(x^3)}{(x^3)^2 + 1} \\
 &= \tan^{-1}\left(x - \frac{1}{x}\right) - \frac{2}{3} \tan^{-1}(x^3) + c
 \end{aligned}$$

7. Ans. (B,C)

Sol. Clearly if $z = c \Rightarrow \rho = 0$ and $t \in \mathbb{R}$
if $z - c \neq 0$.

$$|z - c| = |\rho| \left| \frac{1 + it}{1 - it} \right| \Rightarrow \rho = \pm |z - c|$$

$$\text{Also. } \frac{z - c}{\rho} = \frac{1 + it}{1 - it} \Rightarrow t = i \left(\frac{\rho - z + c}{\rho + z - c} \right)$$

$$\Rightarrow t = i \left(\frac{1 - \frac{z - c}{\rho}}{1 + \frac{z - c}{\rho}} \right) = i \left(\frac{1 - \omega}{1 + \omega} \right)$$

$$\text{where } \omega = \frac{z - c}{\rho} = \frac{z - c}{\pm |z - c|} \Rightarrow |\omega| = 1$$

$$= i \left(\frac{(1 - \omega)(1 + \bar{\omega})}{(1 + \omega)(1 + \bar{\omega})} \right) = i \left(\frac{1 - \omega\bar{\omega} - \omega + \bar{\omega}}{1 + \omega\bar{\omega} + \omega + \bar{\omega}} \right)$$

$$= i \left(\frac{-2i \operatorname{Im}(\omega)}{2 + 2\operatorname{Re}(\omega)} \right) = \frac{\operatorname{Im}(\omega)}{1 + \operatorname{Re}(\omega)}$$

$$\Rightarrow t = \frac{\operatorname{Im}(z - c)}{\pm |z - c| + \operatorname{Re}(z - c)}$$

8. Ans. (B,C,D)

$$\text{Sol. } S_n = \sum_{k=1}^{\infty} k^n r^k$$

$$\frac{d}{dr}(S_n) = \sum_{k=1}^{\infty} k^n \cdot k \cdot r^{k-1} = \frac{1}{r} \sum_{k=1}^{\infty} k^{n+1} r^k$$

$$\Rightarrow \frac{d}{dr}(S_n) = \frac{1}{r} S_{n+1}$$

$$\Rightarrow S_{n+1} = r \frac{d}{dr}(S_n)$$

$$\Rightarrow S_{n+1} = r \frac{d}{dr}(S_n)$$

$$\text{Now. } S_0 = \sum_{k=1}^{\infty} r^k = \frac{r}{1 - r}$$

$$S_1 = r \frac{d}{dr}(S_0) = \frac{r}{(1 - r)^2}$$

$$S_2 = r \frac{d}{dr}(S_1) = r \frac{d}{dr} \left(\frac{r}{(1 - r)^2} \right)$$

$$= r \left[\frac{(1 - r)^2 + 2r(1 - r)}{(1 - r)^4} \right] = \frac{r}{(1 - r)^2} + \frac{2r^2}{(1 - r)^3}$$

$$= \frac{1}{r} [S_0^2 + 2S_0^3]$$

9. Ans. (B,C,D)

Sol. Let events A, B, C correspond to prize being behind selected, opened and remaining door respectively. Let H_B denote event that host opens door B.

Probability that he sticks with his choice

$$= P\left(\frac{A}{H_B}\right) \text{ and he switches } = P\left(\frac{C}{H_B}\right)$$

$$\text{Now. } P(A) = P(B) = P(C) = \frac{1}{3}$$

$$P\left(\frac{H_B}{A}\right) = \frac{1}{2}, P\left(\frac{H_B}{B}\right) = 0, P\left(\frac{H_B}{C}\right) = 1$$

$$P(H_B) = P(A) \cdot P\left(\frac{H_B}{A}\right) + P(B) \cdot P\left(\frac{H_B}{B}\right) + P(C) \cdot P\left(\frac{H_B}{C}\right)$$

$$= \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \times 0 + \frac{1}{3} \times 1 = \frac{1}{2}$$

$$P\left(\frac{A}{H_B}\right) = \frac{P\left(\frac{H_B}{A}\right) \cdot P(A)}{P(H_B)} = \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{1}{2}} = \frac{1}{3}$$

$$P\left(\frac{C}{H_B}\right) = \frac{P(C) \cdot P\left(\frac{H_B}{C}\right)}{P(H_B)} = \frac{2}{3}$$

Clearly it is advantageous to switch.

10. Ans. (A,B,C,D)

Sol. Adding equation (i) and (ii)

$$2z = a + b \Rightarrow z = \frac{a+b}{2}$$

Adding (i) and (iii)

$$2y = a + c \Rightarrow y = \frac{a+c}{2}$$

Adding (ii) and (iii)

$$2x = b + c \Rightarrow x = \frac{b+c}{2}$$

$$\Rightarrow x + y + z = a + b + c.$$

11. Ans. (B,C)

Sol. Tangents to hyperbola $\frac{x^2}{2} - \frac{y^2}{3} = 1$

$$y = mx + \sqrt{2m^2 - 3}$$

passes through (α, β)

$$\Rightarrow (\beta - m\alpha)^2 = 2m^2 - 3$$

$$\Rightarrow m^2(\alpha^2 - 2) - 2\alpha\beta m + \beta^2 + 3 = 0$$

$\tan A \cdot \tan B$ = product of roots of above

$$\text{equation} = \frac{\beta^2 + 3}{\alpha^2 - 2} = k$$

$$\text{if } k = 2, \frac{\beta^2 + 3}{\alpha^2 - 2} = 2 \Rightarrow \beta^2 + 3 = 2\alpha^2 - 4$$

$$\Rightarrow \beta^2 = 2\alpha^2 - 7$$

$$\text{if } k = 3, \beta^2 + 3 = 3\alpha^2 - 6$$

$$\Rightarrow \beta^2 = 3\alpha^2 - 9$$

12. Ans. (B,D)

Sol. $(a+b)^{2n} + (a-b)^{2n} = 2 \sum_{r=0}^n \binom{2n}{2r} a^{2n-2r} b^{2r}$

$$\text{put } a = 1, b = \sqrt{3}$$

$$2 \sum_{r=0}^n \binom{2n}{2r} (\sqrt{3})^{2r} = (\sqrt{3} + 1)^{2n} + (\sqrt{3} - 1)^{2n}$$

$$= (4 + 2\sqrt{3})^n + (4 - 2\sqrt{3})^n$$

$$= 2^n \left((2 + \sqrt{3})^n + (2 - \sqrt{3})^n \right)$$

$$\Rightarrow \sum_{r=0}^n \binom{2n}{2r} (\sqrt{3})^{2r} = 2^{n-1} \left((2 + \sqrt{3})^n + (2 - \sqrt{3})^n \right)$$

$$\Rightarrow T_n = (2 + \sqrt{3})^n + (2 - \sqrt{3})^n$$

$$\Rightarrow T_{n+2} - [(2 + \sqrt{3}) + (2 - \sqrt{3})] T_{n+1} + (2 + \sqrt{3})(2 - \sqrt{3}) T_n = 0$$

$$\Rightarrow T_{n+2} = 4T_{n+1} - T_n$$

$$T_1 = {}^2C_0 + 3 \times {}^2C_2 = 4$$

$$2T_2 = {}^4C_0 + 3 \cdot {}^4C_2 + 3^2 \cdot {}^4C_4$$

$$\Rightarrow 2T_2 = 1 + 18 + 9 \Rightarrow T_2 = 14$$

$$T_3 = 4 \times T_2 - T_1 = 4 \times 14 - 4 = 52$$

$$T_4 = 4 \times T_3 - T_2 = 4 \times 52 - 14 = 194$$

SECTION-II

1. Ans. 0.37

$$\text{Sol. } \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\left(\frac{\tan x}{x - \tan x} \right)}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{\tan x}{x - \tan x} \right) \left(\frac{\tan x - x}{x} \right)} = e^{-\lim_{x \rightarrow 0} \frac{\tan x}{x}} = e^{-1}$$

$$= \frac{1}{e} = 0.37$$

2. Ans. 0.66 or 0.67

$$\text{Sol. } \int_0^1 f(x) dx = \sum_{n=1}^{\infty} \int_{\frac{1}{2^n}}^{\frac{1}{2^{n-1}}} f(x) dx = \sum_{n=1}^{\infty} \int_{\frac{1}{2^n}}^{\frac{1}{2^{n-1}}} \frac{1}{2^{n-1}} dx$$

$$\int_{\frac{1}{2^n}}^{\frac{1}{2^{n-1}}} f(x) dx = \frac{1}{2^{n-1}} \left[\frac{1}{2^{n-1}} - \frac{1}{2^n} \right] = \frac{1}{2^{2n-1}}$$

$$\therefore \int_0^1 f(x) dx = \sum_{n=1}^{\infty} \frac{1}{2^{2n-1}} = \frac{\frac{1}{2}}{1 - \frac{1}{4}} = \frac{2}{3}$$

3. Ans. 1.73

$$\text{Sol. } (\tan^{-1}y - x)dy = (1 + y^2)dx$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{1 + y^2} = \frac{\tan^{-1}y}{1 + y^2}$$

$$IF = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1} y}$$

$$x \cdot e^{\tan^{-1} y} = \int \frac{\tan^{-1} y}{1+y^2} \cdot e^{\tan^{-1} y} dy$$

$$I = \int \frac{\tan^{-1} y}{1+y^2} e^{\tan^{-1} y} dy$$

$$\text{Let } \tan^{-1} y = t$$

$$\frac{1}{1+y^2} dy = dt$$

$$\int t e^t dt = t e^t - e^t + c = e^{\tan^{-1} y} (\tan^{-1} y - 1) + c$$

$$\therefore x \cdot e^{\tan^{-1} y} = e^{\tan^{-1} y} (\tan^{-1} y - 1) + c$$

$$\Rightarrow x = \tan^{-1} y - 1 + c e^{-\tan^{-1} y}$$

$$\text{passes through } (0, \tan 1)$$

$$0 = 1 - 1 + c \Rightarrow c = 0$$

$$\therefore x = \tan^{-1} y - 1 \Rightarrow \tan^{-1} y = (x + 1)$$

$$\Rightarrow y = \tan(x + 1)$$

$$\text{for } x = \frac{\pi}{3} - 1$$

$$y = \tan\left(\frac{\pi}{3}\right) = \sqrt{3} = 1.732$$

4. Ans. 1.50

$$\text{Sol. } \vec{c} \times \vec{a} = \vec{b} \Rightarrow |\vec{c} \times \vec{a}| = |\vec{b}|$$

$$\Rightarrow |\vec{c}| |\vec{a}| \sin \theta = 3 \Rightarrow |\vec{c}| = \frac{3}{2 \sin \theta}$$

$$|\vec{c} - \vec{a}|^2 = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a}$$

$$= |\vec{c}|^2 + 4 - 2|\vec{c}| |\vec{a}| \cos \theta$$

$$= \frac{9}{4} \csc^2 \theta + 4 - 6 \cot \theta$$

$$= \frac{9}{4} + \left(\frac{3}{2} \cot \theta - 2 \right)^2$$

$$\Rightarrow |\vec{c} - \vec{a}|^2 \geq \frac{9}{4} \Rightarrow |\vec{c} - \vec{a}| \geq \frac{3}{2}$$

$$\therefore \text{Least value} = 1.50$$

5. Ans. 4.00

Sol. Given plane can be

$$(2x + 3y - z + 1) + \lambda(x + y - 2z + 3) = 0$$

$$\Rightarrow x(2 + \lambda) + y(3 + \lambda) - z(1 + 2\lambda) + 1 + 3\lambda = 0$$

$$\text{It is perpendicular to } 3x - y - 2z - 4 = 0$$

$$\Rightarrow 3(2 + \lambda) - (3 + \lambda) + 2(1 + 2\lambda) = 0$$

$$\Rightarrow 6\lambda + 5 = 0 \Rightarrow \lambda = -\frac{5}{6}$$

$$\therefore \text{Plane is } 7x + 13y + 4z - 9 = 0$$

$$\therefore \ell = 7 \quad m = 4 \quad n = -9$$

$$\frac{\ell - n}{m} = \frac{16}{4} = 4.00$$

6. Ans. 1.70

Sol. 2, 3, ..., n-1, n.

$$1, 3, \dots, n.$$

$$\dots$$

$$1, 2, 3, \dots, n-1.$$

In each sequence missing number can be accommodated in $(n-1)$ ways, but in every consecutive pair one sequence is recounted.

$$\therefore f(n) = n(n-1) - (n-1) + 1 = n^2 - 2n + 2$$

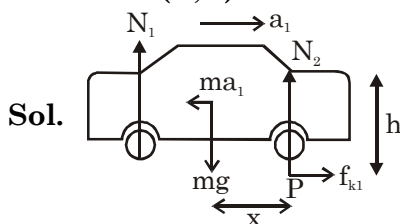
$$f(4) = 16 - 8 + 2 = 17$$

$$f(5) = 25 - 10 + 2 = 17$$

$$\therefore \frac{f(5)}{f(4)} = 1.70$$

PAPER-2**PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (A,C)



Rear wheels are locked :

If frame of reference is car :

$$ma_1 h + mgx = N_1 \times 2x$$

$$N_1 = \frac{ma_1 h + mgx}{2x}$$

$$\& N_1 + N_2 = mg$$

$$N_2 = mg - N_1$$

$$= mg - \frac{ma_1 h}{2x} - \frac{mg}{2}$$

$$N_2 = m \left(\frac{g}{2} - \frac{a_1 h}{2x} \right)$$

Retardation of car,

$$a_1 = \frac{f_{k1}}{m} = \frac{\mu N_2}{m}$$

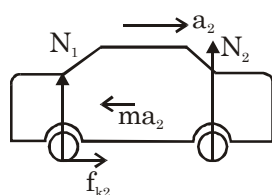
$$a_1 = \mu \left(\frac{g}{2} - \frac{a_1 h}{2x} \right)$$

$$a_1 \left(1 + \frac{\mu h}{2x} \right) = \frac{\mu g}{2}$$

$$a_1 = \frac{\mu g}{2 \left(1 + \frac{\mu h}{2x} \right)} = \frac{\mu g}{2 + \frac{\mu h}{x}} = \frac{g}{\frac{2}{\mu} + \frac{h}{x}}$$

$$L_1 = \frac{v^2}{2a_1} = \frac{v^2}{2g} \left(\frac{2}{\mu} + \frac{h}{x} \right)$$

Front wheels are locked



$$a_2 = \frac{f_{k2}}{m} = \frac{\mu N_1}{m} = \mu \left(\frac{g}{2} + \frac{a_2 h}{2x} \right)$$

$$a_2 \left(1 - \frac{\mu h}{2x} \right) = \frac{\mu g}{2}$$

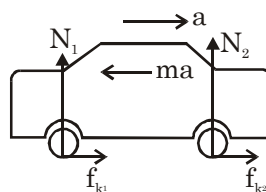
$$a_2 = \frac{\mu g}{2 \left(1 - \frac{\mu h}{2x} \right)}$$

$$a_2 = \frac{g}{\frac{2}{\mu} - \frac{h}{x}}$$

$$L_2 = \frac{v^2}{2a_2} = \frac{v^2}{2g} \left(\frac{2}{\mu} - \frac{h}{x} \right)$$

Thus, $L_1 > L_2$

when all wheels are locked



$$a = \frac{f_{k1} + f_{k2}}{m} = \frac{\mu (N_1 + N_2)}{m}$$

$$a = \mu g$$

$$L = \frac{v^2}{2a} = \frac{v^2}{2\mu g} = \frac{L_1 + L_2}{4}$$

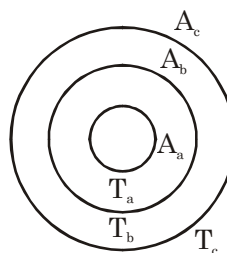
2. Ans. (A,B,D)

Sol. Power leaving the system = P

$$P = \sigma(4\pi C^2)T_c^4$$

Net power leaving B = P

$$P = \sigma(4\pi b^2)T_b^4 - \sigma(4\pi b^2)T_c^4$$



Net power leaving A = P

$$\sigma A_a T_a^4 - \sigma A_b \frac{a^2}{b^2} T_b^4 = P$$

$$\sigma A_a (T_a^4 - T_b^4) = P$$

$$T_a^4 = \frac{P}{\sigma A_a} + T_b^4$$

$$T_a^4 = \frac{P}{4\pi\sigma} \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)$$

If both outer & middle shell were removed

$$\sigma(4\pi a^2) T_a^4 = P$$

$$T_a^4 = \frac{P}{4\pi\sigma a^2}$$

3. Ans. (B,C)

Sol. In case (a)

$$L_{\text{eff}} = L$$

In case (b)

$$L_{\text{eff}} = L + L = 2L$$

In case (c)

$$L_{\text{eff}} = L + L + 2M = 2L + 2M$$

$$\tan \phi = \frac{x_L}{R} = \frac{\omega L_{\text{eff}}}{R}$$

4. Ans. (D)

Sol. $V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\ell T}{m}}$

$$T = \frac{\ell}{V} = \sqrt{\frac{M\ell}{T}}$$

5. Ans. (A,B,D)

Sol. Curve is symmetric near minimum deviation.

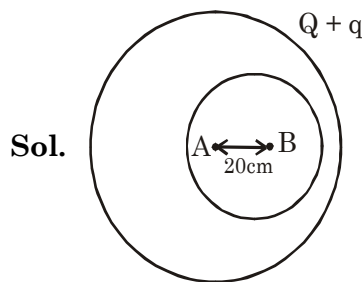
$$\mu = \frac{\sin\left(\frac{\delta_{\min} + A}{2}\right)}{\sin\left(\frac{A}{2}\right)} \Rightarrow \mu = \sqrt{2}$$

Critical angle is 45° . If $r_1 < 15^\circ$ then TIR will happen on second face.

6. Ans. (C, D)

Sol. Since there is no parallax image formed in the same place so $MI = 2f$. We need position of M & I.

7. Ans. (B,D)



Sol.

The charge distribution on the inner surface of the sphere will be non-uniform, while charge distribution on the outer surface will be uniform.

$$\begin{aligned} \therefore V_B &= \frac{k(Q+q)}{R} + \frac{k(-q)}{r} + \frac{kq}{r'} \\ &= \frac{K \times 9 \times 10^{-6}}{1} + \frac{k(-5) \times 10^{-6}}{0.5} + \frac{k(5 \times 10^{-6})}{0.2} \\ &= 216 \text{ kV} \end{aligned}$$

Also outside the sphere field is due to charge on the outer surface which does not change on changing the position of charge in the cavity. Hence field outside the sphere will not change.

8. Ans. (A,B,D)

Sol. Just before point A,

$$N_1 = mg$$

Just after point A,

$$N_2 - mg = \frac{mv^2}{R} \Rightarrow N_2 = mg + \frac{mv^2}{R}$$

$$\therefore N_1 = 40 \text{ \& } N_2 = 40 + \frac{4 \times 16}{1} = 104$$

Applying work energy theorem,

$$-mgh = \frac{1}{2}mv'^2 - \frac{1}{2}mv^2$$

$$\Rightarrow -4 \times 10 \times \frac{1}{2} = \frac{1}{2} \times 4 \times V'^2 - \frac{1}{2} \times 4 \times 16$$

$$\therefore V' = \sqrt{6} \text{ m/s}$$

$$\text{From geometry, } R - R\cos\theta = \frac{R}{2}$$

$$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

Note : At maximum height, horizontal component of velocity will remain non-zero.

9. **Ans. (A)**

Sol. θ is small

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

Tension when string is vertical

$$= mg + m\ell\Omega^2$$

$$= mg + m\ell(\theta_0\omega)^2$$

$$= mg + m\ell\left(\theta_0\sqrt{\frac{g}{\ell}}\right)^2$$

$$= mg(1 + \theta_0^2)$$

Kinetic energy at mean position

$$k = \frac{1}{2}I\Omega^2$$

$$k = \frac{1}{2}(m\ell^2)\left(\theta_0\sqrt{\frac{g}{\ell}}\right)^2 = \frac{1}{2}m\ell\theta_0^2g$$

Angular acceleration at extreme position

$$\alpha = \omega^2\theta_0$$

$$\alpha = \frac{g}{\ell}\theta_0$$

10. **Ans. (C)**

Sol. θ is small acceleration at mean position

$$a = \ell\Omega^2 = \ell(\theta_0\omega)^2$$

$$= \ell\theta_0^2 \times \frac{g}{\ell} = \theta_0^2g$$

$$\text{acceleration at extreme position, } a = \ell\alpha \\ = \ell(\omega^2\theta_0)$$

$$= \ell\left(\frac{g}{\ell}\theta_0\right) = \theta_0g$$

Tension at extreme position

$$T = mg \cos \theta$$

$T = mg$ when θ is small

Mechanical energy of oscillation

= K.E. at mean position

$$= \frac{1}{2}mv^2 = \frac{1}{2}m(\ell\Omega)^2 = \frac{1}{2}m\ell^2(\theta_0\omega)^2$$

$$= \frac{1}{2}m\ell^2\theta_0^2 \times \frac{g}{\ell} = \frac{1}{2}m\ell\theta_0^2g$$

11. **Ans. (C)**

Sol. Intensity of light is given by $I = \frac{\dot{n}hC}{A\lambda}$

Also when position of jockey are interchanged, polarity of the plates changes.

12. **Ans. (D)**

Sol. Since $V_0 \gg \phi$ (work function)

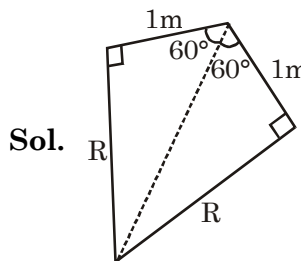
$\Rightarrow$ all the electrons ejected due to PEE reaches the +ve plate.

On increasing V_0 , K.E. of is reaching at +ve plate increases thus decreasing the cut off wavelength of X-rays.

Changing the plate material will change atomic no. thus nature of characteristic X-ray.

SECTION-II

1. **Ans. 1.10 to 1.15**



Sol.

$$\frac{\pi}{3} = \omega t$$

$$\tan 60 = \frac{R}{1} = \sqrt{3} = \frac{mv}{qB}$$

$$B = \frac{9 \times 10^{-31}}{1.6 \times 10^{-19}} \times \frac{\sqrt{3}}{2} = \frac{9}{3.2} \times 10^{-12}$$

$$= 4\pi \times 10^{-5} \times 200 \text{ i}$$

$$i = \frac{9}{3.2 \times 8\pi} \times 10^{-7}$$

2. **Ans. 5.00**

Sol. $v \cos \theta = 5$

$$v \cos(53^\circ - \theta) = 3$$

$$v \left[\frac{3}{5} \cos \theta + \frac{4}{5} \sin \theta \right] = 3$$

$$\frac{4}{5} v \sin \theta = 0$$

$$\theta = 0^\circ$$

$$\Rightarrow v = 5$$

3. **Ans. 960.00**

Sol. Net work done is zero.

$$(\rho - 800) \times gV \times 1.5 = (1200 - \rho)gV \times 1$$

$$\rho = \frac{4800}{5} = 960 \text{ kg/m}^3$$

4. **Ans. 22.22**

$$\text{Sol. } \lambda_1 = \frac{h}{m(v_1 - v_2)} = 25\mu$$

$$\lambda_2 = \frac{h}{m(v_1 + v_2)} = 20\mu\text{m}$$

$$\Rightarrow \frac{v_1 + v_2}{v_1 - v_2} = \frac{5}{4} \Rightarrow v_2 = \frac{v_1}{9}$$

again

$$\lambda_1 = \frac{h}{m\left(v_1 - \frac{v_1}{9}\right)} = 25\mu\text{m} \Rightarrow \frac{h}{mv_1} = \frac{25 \times 8}{9} \mu\text{m}$$

$$\lambda = \frac{h}{mv_1} = \frac{200}{9} \mu\text{m} = 22.22\mu\text{m}$$

5. Ans. 0.40

Sol. $-1 \times 0.04 \times 4 + \frac{1}{2} \times 2 \times v^2 = 0$

6. Ans. 7.35 to 7.41

Sol. _____

$$v = v_0 e^{-t/RC} = 20e^{-\frac{t}{2}}$$

$$qE = mg$$

$$q \times \frac{20}{d} = mg$$

$$mg - qE = ma$$

$$mg(1 - e^{-t/2}) = ma = \frac{dv}{dt}$$

$$v = 10 [t + 2e^{-t/2} - 2]$$

$$v = \frac{20}{e} \text{ at } t = 2$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans.(B)
2. Ans.(B)
3. Ans.(D)
4. Ans.(A,B,C)

For liquid $\rightarrow$ gas, ΔH_{sys} + ve, ΔS_{sys} + ve

As it is equilibrium condition, process is thermodynamically reversible and hence, $\Delta G_{\text{sys}} = 0$, $\Delta S_{\text{univ}} = 0$

5. Ans.(A,B,D)

$$Z_A = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$$

$$Z_B = 1 + 12 \times \frac{1}{4} = 4$$

Hence, simplest formula is $A_4B_4 = AB$, and there is 4 AB formula units per unit cell.

As 'B' atoms are smaller than 'A' atoms,

they are not in contact but as $\frac{r_B}{r_A} = (\sqrt{2} - 1)$

= ideal value of octahedral void, 'B' atoms are in contact.

6. Ans.(A,C,D)
7. Ans.(C)
8. Ans.(A,C,D)
9. Ans.(D)
10. Ans.(B)
11. Ans.(A)
12. Ans.(C)

SOLUTION

SECTION-II

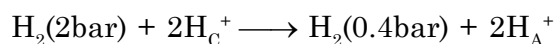
1. Ans.(5.50)

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

$$\text{or, } (35.5 + \frac{a \times 5^2}{5^2})(5 - 5 \times 0.1) = 5 \times 0.082 \times 450$$

$$\therefore a = 5.50 \text{ atm L}^2 \text{ mol}^{-2}$$

2. Ans.(-0.90)



$$Q = \frac{0.4}{2} \times \frac{2 \times 10^{-5} \times 0.2}{4 \times 10^{-6} \times 0.1} = 2$$

$$E_{\text{cell}} = \frac{-0.06}{2} \log 2 = -0.03 \times 0.3$$

$$= -0.009$$

$$x \times 10^{-2} = (-0.9) \times 10^{-2}$$

$$= (-0.9)$$

3. Ans.(1650.00)

$$\Delta t = \frac{t_{1/2}}{\ln 2} \cdot \ln \frac{r_A}{r_B} = \frac{5775}{\ln 2} \times \ln \frac{5000}{4000} = 1650 \text{ yrs.}$$

4. Ans.(5.00)

Angular nodes = $2 = l \Rightarrow d$ - orbital $\Rightarrow$ five in number.

5. Ans.(3.00)

6. Ans.(3.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,C)**

$$\operatorname{cosec}\left(\frac{\pi}{4} + x\right) + \operatorname{cosec}\left(\frac{\pi}{4} - x\right) = 2\sqrt{2}$$

$$\frac{\sqrt{2}}{\sin x + \cos x} + \frac{\sqrt{2}}{\cos x - \sin x} = 2\sqrt{2}$$

$$\frac{2 \cos x}{\cos^2 x - \sin^2 x} = 2$$

$$\cos 2x = \cos x$$

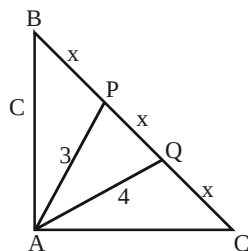
$$2 \sin \frac{3x}{2} \cdot \sin \frac{x}{2} = 0$$

$$x = 2n\pi, \frac{2n\pi}{3}$$

$$x \in [0, 4\pi] \text{ is } 7$$

$$x \in (0, 2\pi) \text{ is } 2$$

2. **Ans. (C)**



$$9 = c^2 + x^2 - 2cx \cos B$$

$$\cos B = \frac{c}{3x}$$

$$9 = c^2 + x^2 - 2cx \left(\frac{c}{3x}\right) = \frac{c^2}{3} + x^2 \quad \dots(1)$$

Similarly in $\triangle ACQ$

$$16 = \frac{b^2}{3} + x^2 \quad \dots(2)$$

Adding (1) + (2)

We get

$$25 = \left(\frac{b^2 + c^2}{3}\right) + 2x^2$$

$$25 = 5x^2, (b^2 + c^2 = a^2 = 9x^2) \Rightarrow x = \sqrt{5}$$

$$\ell(BC) = 3\sqrt{5}$$

3. **Ans. (A,B,D)**

$$ty = x + 2t^2 \xrightarrow{(6,8)} 8t = 6t + 2t^2$$

$$t^2 - 4t + 3 = 0 \Rightarrow t = 1, 3 \quad (2t^2, 4t) \equiv Q \text{ \& R}$$

$$Q = (2, 4) \quad P = (6, 8)$$

$$R \equiv (18, 12)$$

$$\text{area} = \frac{(64 - 48)^{3/2}}{4} = \frac{64}{4} = 16$$

$$PS : 2x - y - 4 = 0$$

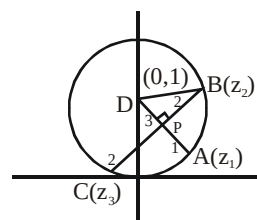
Equation of circle circumscribing the $\triangle PQR$ is normal at Q and R intersect at $(30, -24) = S$

Equation of circle on PS as diameter

$$(x - 30)(x - 6) + (y + 24)(y - 8) = 0$$

$$x^2 + y^2 - 36x + 16y - 12 = 0$$

4. **Ans. (A,C)**



$$\frac{3z_1 + i}{4} = \frac{2z_2 + 2z_3}{4}$$

$$AD = 1$$

$$DP = \frac{3}{4}$$

$$r^2 = BP^2 + DP^2 \Rightarrow BP = 1 - \frac{9}{16} = \frac{\sqrt{7}}{4}$$

$$BC = \frac{\sqrt{7}}{2}$$

$$\text{Area of quad } ABCD = \frac{1}{2} \times \frac{\sqrt{7}}{2} \times 1 = \frac{\sqrt{7}}{4}$$

5. **Ans. (A,C)**

$$\text{Let } S = \sum_{k=1}^{\infty} \frac{P_k}{3^k} = \frac{P_1}{3} + \frac{P_2}{3^2} + \frac{P_3}{3^3} + \frac{P_4}{3^4} + \frac{P_5}{3^5}$$

$$S = \frac{1}{3} + \frac{1}{3^2} + \frac{2}{3^3} + \frac{3}{3^4} + \frac{5}{3^5} + \frac{8}{3^6} + \dots + \frac{P_n}{3^n}$$

$$\frac{S}{3} = \frac{1}{3^2} + \frac{1}{3^3} + \frac{2}{3^4} + \frac{3}{3^5} + \dots + \frac{P_{n-1}}{3^n} + \dots$$

$$\frac{S}{9} = \frac{1}{3^3} + \frac{1}{3^4} + \frac{2}{3^5} + \dots + \frac{P_{n-2}}{3^n}$$

$$S - \left(\frac{S}{3} + \frac{S}{9}\right) = \frac{1}{3} \Rightarrow S = \frac{3}{5}$$

6. Ans. (B,C,D)

Let H = event that married man watches the show
W = event that married women watches the show

$$P(H) = 0.4, \quad P(W) = 0.5, \quad P\left(\frac{H}{W}\right) = 0.7$$

$$P(H \cup W) = 0.35$$

$$P\left(\frac{W}{H}\right) = \frac{0.35}{0.4} = \frac{7}{8}, \quad P(H \cup W) = 0.55$$

7. Ans. (A,C)

Directly expand put $-x = t$

$$\Delta = t(t^2 - ab) - a(bt - a^2) + b(b^2 - at)$$

$$\Delta = t^3 + a^3 + b^3 - 3abt = (t + a + b)$$

$$(t + a + b)(t^2 + a^2 + b^2 - at - bt - ab)$$

$$(a + b - x)(x^2 + x(a + b) + a^2 + b^2 - ab)$$

$$\text{If } a = b \quad \Delta = (2a - x)(x^2 + 2ax + a^2)$$

8. Ans. (B,C)

$$A \cdot \text{adj}(2B) = 16I$$

$$A(4 \text{adj} B) = 16I \Rightarrow A \cdot \text{adj} B = 4I$$

$$\Rightarrow A | B | B^{-1} = 4I$$

$$AB^{-1} = 2I \Rightarrow A = 2B$$

$$B \cdot \text{adj} A = B \cdot \text{adj}(2B) = 4B \cdot \text{adj} B = 4 | B | I_3 = 8I_3$$

$$A \cdot \text{adj} B = 4I \Rightarrow A^{-1} A \cdot \text{adj} B$$

$$\Rightarrow \text{adj} B = 4I \Rightarrow A^{-1} \cdot A \cdot \text{adj} B = 4A^{-1}$$

$$\Rightarrow \text{adj} B = 4A^{-1}$$

$$A^{-1} \cdot \text{adj} B = A^{-1} \cdot (4A^{-1}) = 4(A^{-1})^2$$

$$(A^{-1}(\text{adj} B))^{-1} = (4A^{-1}A^{-1})^{-1}$$

$$= \frac{1}{4} A^2 = \frac{1}{4} (2B)^2 = B^2$$

Solution Q.9 & Q. 10

9. Ans. (A)

10. Ans. (B)

$$I = \int_{e^{-x}}^1 (-\ln t)^n dt$$

$$-\ln t = u \Rightarrow t = e^{-u} \Rightarrow dt = -e^{-u} du$$

$$I = \int_0^x u^n e^{-u} du$$

Hence,

$$I_n = \int_0^x \underbrace{u^n}_{\text{I}} \underbrace{e^{-u}}_{\text{II}} du = \left| -u^n e^{-u} \right|_0^x + \int_0^x n u^{n-1} e^{-u} du$$

$$\Rightarrow I_n = 0 + n I_{n-1} \Rightarrow I_n = n(n-1) I_{n-2}$$

$$\therefore I_n = n(n-1)(n-2) \dots 2 \cdot I_1$$

$$\Rightarrow I_1 = \int_0^\infty u e^{-u} du = 1$$

$$\Rightarrow I_n = n!$$

Let

$$J = \int_0^{1/2} \left(\frac{1}{4} - x^2 \right)^4 dx = \int_0^{1/2} \left(\frac{1}{4} - \left(\frac{1}{2} - x \right)^2 \right)^4 dx$$

(Using King property)

$$\therefore J = \int_0^{1/2} (x - x^2)^4 dx = \int_0^{1/2} x^4 (1 - x)^4 dx \dots (i)$$

$$\Rightarrow J = K, \text{ So } J - K = 0$$

$$\text{Put } x = 1 - y$$

$$\text{Also, } J = - \int_1^{1/2} (1 - y)^4 y^4 dy$$

$$\Rightarrow J = \int_{1/2}^1 (1 - x)^4 x^4 dx \dots (ii)$$

$$\therefore (i) + (ii)$$

$$\Rightarrow J = \frac{1}{2} \int_0^1 x^4 (1 - x)^4 dx$$

(Using Queen property)

$$\text{Put } x = \sin^2 \theta,$$

$$J = \int_0^{\pi/2} \sin^9 \theta \cdot \cos^9 \theta d\theta$$

$$= \frac{(8 \times 6 \times 4 \times 2)(8 \times 6 \times 4 \times 2)}{(18 \times 16 \times 14 \times 12 \times 10 \times 8 \times 6 \times 4 \times 2)} = \frac{1}{1260}$$

Solution Q.11 & Q. 12

11. Ans. (A)

12. Ans. (B)

$$L_2 : x - 3y - 4 = 0 \text{ \& } 4y - z + 5 = 0$$

$$P \equiv (4, 0, 5) \quad \overrightarrow{DR} = \vec{n} = \vec{n}_1 \times \vec{n}_2$$

$$\vec{n} = 3\hat{i} + \hat{j} + 4\hat{k} \quad \begin{pmatrix} \vec{n}_1 = \hat{i} - 3\hat{j} \\ \vec{n}_2 = 4\hat{j} - \hat{k} \end{pmatrix}$$

$$L_2 : \frac{x-4}{3} = \frac{y}{1} = \frac{z-5}{4} \dots (1)$$

$$P_1 : x + y - z = 5, \quad L_2 : (3\lambda + 4, \lambda, 4\lambda + 5)$$

$$\frac{x - (3\lambda + 4)}{1} = \frac{y - \lambda}{1} = \frac{z - (4\lambda + 5)}{-1} = -2 \left(\frac{-6}{3} \right) = 4$$

$$(x, y, z) \equiv (3\lambda + 8, \lambda + 4, 4\lambda + 1)$$

$$\text{Image of line } L_2 : \frac{x-8}{3} = \frac{y-4}{1} = \frac{z-1}{4}$$

$$P_1 = 0; P_3 = 0; P_1 + \lambda P_3 = 0$$

$$(x + y - z - 5) + \lambda(x - 3y - 4) = 0$$

$$x(1 + \lambda) + y(1 - 3\lambda) - z - (5 + 4\lambda) = 0$$

$$p_1 = p_2 \text{ from } (4, 0, 0)$$

$$\Rightarrow \left| \frac{1}{\sqrt{(\lambda+1)^2 + (1-3\lambda)^2 + 1}} \right| = \left| \frac{1}{\sqrt{3}} \right|$$

$$\lambda = 0, \frac{2}{5}$$

$$P : 7x - y - 5z - 33 = 0$$

$$L_1 : x + y - z = 5, 2x - y + \lambda z = 3$$

$$\text{Point} \equiv \left(\frac{8}{3}, \frac{7}{3}, 0 \right)$$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 2 & -1 & \lambda \end{vmatrix} = (\lambda - 1)\hat{i} - (\lambda + 2)\hat{j} - 3\hat{k}$$

If line L_1 and L_2 are coplanar

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -7 & 5 \\ 3 & 1 & 4 \\ (\lambda - 1) & -(\lambda + 2) & -3 \end{vmatrix} = 0$$

$$\lambda = -\frac{5}{4}$$

SECTION-II

1. **Ans. 4.00**

$-2 \leq f'(x) \leq 2$, Apply LMVT in $x \in [1, 2]$ and $x \in [2, 4]$

$$f(2) - f(1) = f'(x) \Rightarrow f(2) - 2 \in [-2, 2]$$

$$f(2) \in [0, 4] \quad \dots(1)$$

$$\frac{f(4) - f(2)}{2} \in [-2, 2] \Rightarrow f(2) - f(4) \in [-4, 4]$$

$$f(2) \in [4, 12] \quad \dots(2)$$

So from (1) and (2), $f(2) = 4$

2. **Ans. 5.00**

$$f'(x) = \frac{(4a - 2x)}{(4ax - x^2) \log a} \geq 0$$

$$f'(x) = \frac{(x - 2a)}{x(x - 4a) \log a} \geq 0$$

$$\text{C-1 : } a \in (1, \infty) \Rightarrow \frac{(x - 2a)}{x(x - 4a)} \geq 0$$

$$\begin{array}{ccccccc} & - & & + & & - & & + \\ & | & & | & & | & & | \\ & 0 & & 2a & & 4a & & \end{array}$$

$x \in (0, 2a)$ so always $\uparrow$

C-2 : $a \in (0, 1)$

$$\frac{(x - 2a)}{x(x - 4a)} < 0$$

$$\begin{array}{ccccccc} & - & & + & & - & & + \\ & | & & | & & | & & | \\ & 0 & & 2a & & 4a & & \end{array}$$

$x \in (2a, 4a)$

$$2a \leq \frac{3}{2} \Rightarrow a \leq \frac{3}{4} \text{ and } 4a > 2 \Rightarrow a > \frac{1}{2}$$

$$\text{So } a \in \left(\frac{1}{2}, \frac{3}{4} \right] \cup [1, \infty)$$

3. **Ans. 15.00**

$$I = \int \frac{(\sec x - \tan x)}{\sqrt{\sin^2 x - \sin x}} dx$$

$$I = \int \frac{(\sec x - \tan x) \sec x}{\sqrt{\tan^2 x - \sec x \tan x}} dx$$

Put $-\sec x + \tan x = t$

$$= -\sec x \tan x + \sec^2 x = \frac{dt}{dx}$$

$$\text{also } (\tan x - \sec x)^2 = t^2$$

$$2\tan^2 x - 2\tan x \sec x = t^2 - 1$$

$$I = \int \frac{dt}{\sqrt{t^2 - 1}} = \sqrt{2} \int \frac{dt}{\sqrt{t^2 - 1}}$$

$$I = \sqrt{2} \log_e |t + \sqrt{t^2 - 1}|$$

$$I = \sqrt{2} \log_e |(\tan x - \sec x) + \sqrt{2} \sqrt{\tan^2 x - \tan x \sec x}| + c$$

$$k = \sqrt{2}$$

$$f\left(\frac{4\pi}{3}\right) = \sqrt{3} + 2$$

4. Ans. 5.00

$$dy = \frac{(1-x)}{e^x} dx \quad \text{Integrate}$$

$$y = x e^{-x} + c, f(0) = 0 \Rightarrow c = 0$$

$$y = x e^{-x}$$

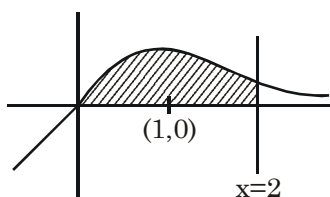
$$\frac{d^2 y}{dx^2} = (x-2)e^{-x}$$

$$\text{Point of inflection} \equiv \left(2, \frac{2}{e^2}\right)$$

$$A = \int_0^2 x e^{-x} dx$$

$$A = 1 - \frac{3}{e^2}$$

$$p + q = 5$$



5. Ans. 7.00

$$(x-c)^2 + (y-c)^2 = c^2, \quad c > 0$$

$$p = r, \quad \left| \frac{7c-12}{5} \right| = c \Rightarrow c = 1, 6$$

$$\text{Sum} = 7$$

6. Ans. 6.00

$$f'(0^+) = \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h}$$

$$f'(0^+) = \lim_{h \rightarrow 0^+} \frac{h^{p-2} \sin \frac{1}{h} + h |\tanh|^{q-3}}{h}$$

$$f'(0^+) = \lim_{h \rightarrow 0^+} h^{p-3} \sin \frac{1}{h} + (\tanh)^{q-3}$$

It exist only when $p > 3$ and $q \geq 3$, $p + q > 6$

$$[p + q]_{\min} = 6$$